Energy-Efficient Sequential Circuit Design Using Optimized Clock-Gated Johnson Counter for Low-Power VLSI Applications

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Abstract- The modern design of VLSI has primarily focused on the minimization of power dissipation. The dimensions of chips are decreasing, and reliability enhancement through advancements in microelectronics means low power design is no longer a choice but a requirement. Since computer systems rely vastly on sequential circuits, it is quite important that these circuits are designed to have less power dissipation. This paper brings out a low-power design methodology for a Johnson Counter that incorporates a mechanism using clock gating. Thus, the SPICE analysis of power verifies that proposed design leads to less power dissipation as well as lesser interconnections than the conventional designs.

Keywords: clock gated, Johnson Counter, VLSI design, power dissipation, sequential circuits, VLSI Application etc.

I. Introduction

Low-power circuit design has become a crucial focus in modern VLSI design. In the early stages of VLSI technology, major concern was not a power dissipation. However, as feature sizes decrease and clock frequencies increase, power management has become a top priority for system designers [6]. Extensive research is now underway to achieve low-power designs for various systems.

In computer systems, ALU and memory operations are operated in sync with clock sync, making efficient sequential circuit design essential for optimal performance. Counters and registers, key examples of sequential circuits, play an important role in generating synchronized data sequences needed in logic designs. The Johnson Counter, a specific type of counter, is particularly useful for generating unique data sequences and is integral in applications like D/A converters. This paper proposes a Energy Efficient Johnson Counter design using a clock gated approach.

In VLSI circuits, it is important to maintain the power dissipation by using clock gating through effective methods. Since sequential circuits significantly contribute to system power usage, designers are highly focused on reducing their power consumption [7]. Clock gating reduces power dissipation by removing not required clock pulses across systems. Studies show that clock transitions account for a considerable portion of power dissipation (around 15-45%) [8]. Thus, eliminating redundant clock pulses can substantially lower power usage [9]-[10]. Effective clock maintaining or clock gated is essential for low-power sequential circuit designs, and the following section explores this approach in more detail.

In this research, proposing a energy-efficient Johnson Counter design utilizing a clock gating system. The clock gating design includes combinational logic, primarily using gates like XOR, NAND and inverters. Although this approach introduces additional circuitry, SPICE analysis reveals that design achieves lower short-circuit power dissipation compared to traditional designs. Furthermore, proposed design offers a simpler and more organized interconnection layout than conventional designs.

The primary objectives of this research are:

1. Design an optimized Johnson counter that reduces unnecessary power consumption in sequential circuits.

2. Utilize clock gating to selectively enable the clock signal when essential, thereby decreasing dissipation of dynamic power.

3. Reduce redundant clock transitions, thereby lowering power consumption while maintaining circuit performance.

4. Conduct SPICE simulations and compare the proposed design against conventional Johnson counters in terms of dynamic power reduction, static power efficiency, and clock activity.

5. Provide an energy-efficient, low-complexity counter design suitable for modern low-power VLSI applications.

II. Literature Survey

The design of low-power Johnson counters can be greatly enhanced through the application of clock gating, a technique that conserves power by controlling the clock signal delivery to specific parts of the circuit. Taraate (2021) discusses the foundations of ASIC design and RTL development using Verilog, highlighting the importance of RTL-level optimizations for achieving efficient power management in ASIC design [1]. These optimizations form the backbone of low-power design strategies in digital systems, particularly in sequential circuits like Johnson counters.

Loh, Tan, and Sim (2021) emphasize the growing demand in the industry for VLSI design training with commercial EDA tools to address power and efficiency challenges [2]. Their work outlines a comprehensive VLSI design methodology, starting logic synthesis and physical designing, which supports the integration of clock gating techniques at various design stages. The high-level principles for low-power IC design outlined by Yeap, Isa, and Loh (2020) further enhance the approach to integrated circuit design with a focus on energy-efficient practices [3].

Verification is critical for ensuring the reliability of low-power designs, especially in SoC and ASIC designs that incorporate power-saving techniques like clock gating. Mehta (2018) provides a structured approach to functional verification for ASICs and SoCs, emphasizing the need for thorough verification in low-power designs to ensure functionality across various power states [4]. Adir et al. contribute to this effort by proposing a methodology that integrates pre-silicon, post-silicon verification, which is essential for verifying the efficacy of clock gating in power-efficient circuits [5].

Weste, Harris, and Banerjee (2006) provide foundational knowledge on CMOS VLSI design, which is essential for implementing power-saving practices, including clock gating, in CMOS technology [6]. Their book serves as a critical resource for understanding the circuit-level and architectural optimizations necessary for low-power digital design.

Wu, Pedram, and Wu (2000) explore the specific application of clock gating in sequential circuits, illustrating how it reduces power usage by deactivating parts of the circuit not in immediate use [7]. Pedram's comprehensive work on power reduction in Integrated Circuit Design outlines essential strategies and usage for minimizing power across various levels of IC design [8]. Unger's 1981 study on double-edge-triggered flip- flops contributes to this field by introducing techniques that minimize power by limiting clock activation to necessary intervals, a key concept in reducing dynamic power in sequential circuits [9].

The work of Macii, Pedram, and Somenzi (1998) provides essential strategies for early-stage design power prediction, which is crucial in the design flow of low-power ASICs [10]. Wu, Wei, and Pedram's (1998) design of double-edge- triggered flip-flops offers further advancements in flip-flop efficiency, essential for low-power clock-gated circuits [11]. Finally, Wu and Pedram (2000) present an improved approach to low-power sequential circuit design using priority encoding and clock gating, showcasing how optimized clock distribution can significantly lower power usage in Johnson counters and similar sequential designs [12].

In summary, the literature highlights a holistic approach to power-efficient design, from high-level ASIC design method- ologies and verification to specific techniques such as clock gating and double-edge triggering, which all contribute to the effective design of low-power Johnson counters.

III. POWER DISSIPATION USING CLOCK GATED

A. *CMOS Devices Power Dissipation*

CMOS devices have high power efficiency, as they consume almost no power when in the idle or inactive state. Initially, designers considered it a negligible problem regarding power consumption. As of now, because of reduced sizes of chips and increasing speed of devices, it is an important problem for the designers of circuits.

The CMOS devices have two kinds of dissipations:

Static power dissipation in CMOS devices occurs due to sub- threshold leakage currents in nonconducting transistors, gate oxide tunneling currents, reverse-biased diode leakage, and contention currents. Dynamic power dissipation is primarily occurred due to discharge of load capacitance and short-circuit currents during switching. Short-circuited current dissipation has a more critical role in sequential circuits since it occurs for each transition in the clock pulse.

B. *Short Circuit Power Dissipation Switching Mode*

Sequential circuit systems are driven by clock pulse transitions. In double-edge-triggered systems, all actions can be carried out on either the clock's positive or negative edge, or both edges [6][9]. However, because of short-circuit current, these identical clock transitions are one of the major causes of power dissipation in CMOS devices. Figure 1 shows an example of a simple CMOS inverter, and Figure 2 shows the outcomes of its simulation. Fig. 1 depicts the schematic of a simple CMOS inverter. Driving the NMOS and PMOS gates with logic 1, the NMOS becomes ON while the PMOS is OFF, leading to an output logic 0. In the converse case, logic 0 at the gates makes the PMOS ON while the NMOS is OFF, leading to an output logic 1

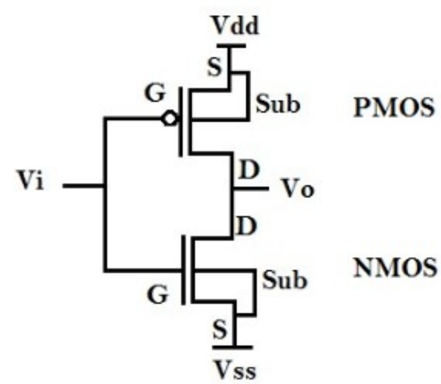

Fig. 1: CMOS Inverter.

During every clock pulse, the transitions from positive to negative edges absorb rise and fall times which, although small, are not zero (for example, when the clock period is 1 µs, the rise and fall times are in the order of 0.01 µs). Sometimes, during these transition periods, NMOS and PMOS transistors turn on momentarily at the same time, resulting in a short circuit between VDD and ground. It results in relatively brief but strong current flow, which dissipates power in the short circuit. The system experiences short-circuit power dissipation with every signal transfer.

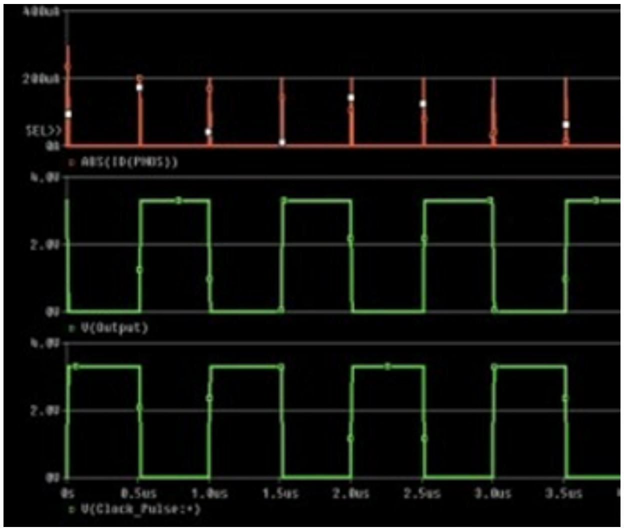

Fig. 2: Short Signal Current at every signal transition.

As indicated in Fig. 2, short-circuit dissipation arising due to clock pulse transitions are of high magnitude. As presented above, a typical value of current rated in a nanoampere scale (nA), current in CMOS microdevices is highly found to show current spiking in microamperes (μA) in the transitions shown in the simulation in Fig. 2 and causes significant loss of power in it. That spiking of current while being described as switching mode short-circuit current for being experienced due to clock transition switching.

C. The Clock Gating Scheme's Function in Lowering Power Dissipation

One of the best ways to lower power dissipation in VLSI systems, particularly in sequential circuits, is clock gating. Clock gating suppresses the undesirable toggling impact of the clock transition by only toggling it when necessary for the circuit. Fig. 3 shows a simple clock gate. Clock gate circuits are easily implemented by the digital designer using combinational logic. The fundamental method is one in which the clock is not directly connected to the flip-flops' non-inverting input. Rather, they are connected by a clock gating mechanism, which is an intermediary block. It clocks every device, allowing only pulses when it wants to perform an action, like toggling or data shifting.
The clock signal is presented to the device when the condition of the logic is true and blocked at that moment when it is false. Clock gating is one of the significant factors in power dissipation where only necessary pulses are made available and unwanted transitions avoided.

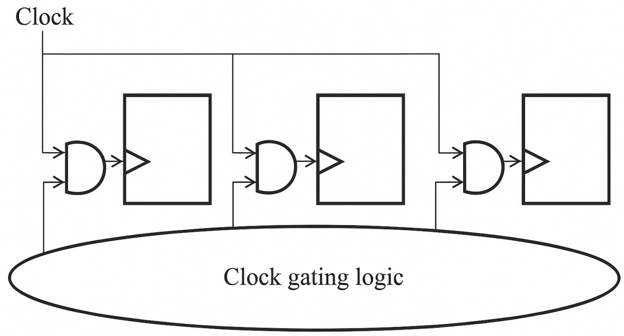

Fig 3. Clock Gating Logic

In the proposed Johnson Counter design, employed the technique of clock gating wherein every J-K flip-flop will receive clock pulses only when required for a toggle on the data. In this design, considered mainly the minimization of short circuit that has formed a dominant factor in the overall dissipation of power caused by transitions on the clock.

IV. CONVENTIONAL DESIGN

Below verilog code shows the implementation of a conventional Johnson Counter as well as the schematic generated in Fig. 4

Initialize Module: Define a module named johnson_counter with inputs clk (clock) and reset, and an output q (4-bit register).
On Every Clock Edge:
If reset is active (high):
Set q to 4'b0000 (reset all bits to 0).
Else:
Perform a left shift on q.
The new least significant bit (q[0]) is the inverted value of the most significant bit (q[3]).
Repeat the process on each positive edge of clk.

A 4-bit Johnson counter, sometimes referred to as a twisted ring counter, is implemented here. It cycles through eight distinct states before repeating.

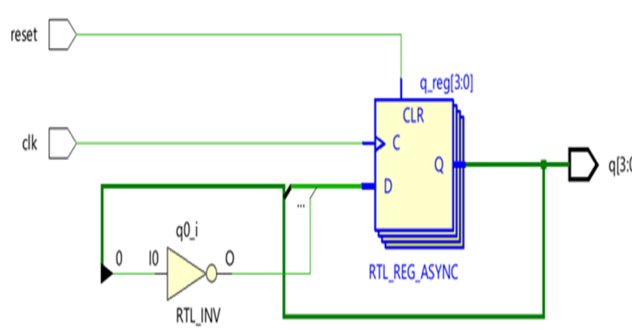

Fig 4. Johnson Counter RTL

V. Proposed Design

In this paper, an optimized implementation of a Johnson counter including clock gating to minimize the dynamic power consumption in sequential circuits is proposed. A Johnson counter is a digital circuit that has plenty of uses. It produces a sequence of bit patterns that has a very important characteristics sequence is Gray code-like, by that it mean that each successive state differs from the previous one in only one bit. However, Johnson counters remain operational and consume current even when in an idle or worthless states, such as 0000 or 1111, at which no useful transitions occur.

A. Clock Gating for Power Efficiency

For power efficiency, try to incorporate clock gating into the Johnson counter's architecture. The process of turning off the clock signal for portions of the circuit when it is not changing is known as clock gating. This lowers dynamic power usage and switching activity. In proposed design, toggling of the clock is allowed only when the counter is in a valid state of the Johnson sequence. Hence, when the counter goes into the invalid state, which is 0000 or 1111, it inhibits the clock signal and no further transitions are allowed until the system has been reset. This saves unnecessary power consumption also and does not allow the counter to stay in an inactive state; rather, it automatically leaves the invalid states.

B. Design and Functionality

The modification core element consists of a 4-bit shift register with its logic cycling through a predefined sequence. The operating functionality of the counter is as that of a standard Johnson counter, but it includes added clock gating logic that monitors the state of the counter. Since the counter transitions into an invalid state, changes stopped through the disconnection of the clock. In case the counter enters an illegal state (0000 or 1111), it gets reset to a legal state automatically, thus always staying in the correct sequence. The schematic for the Johnson Counter with clock gating is shown in Fig. 5.

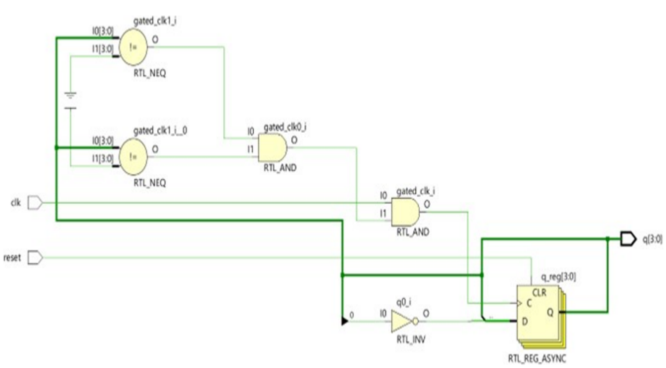

Fig. 5: Johnson Counter with Clock Gating Schematic

The clock gating is controlled through an enable signal that determines what the current state of the counter is. For valid states-that is, those within the normal Johnson counter sequence-the enable signal is asserted and the clock propagates to update the counter. In the invalid states, the enable signal is disserted, thereby stopping the propagation of the clock and maximally saving power.

VI. Results and Discussion

In this section, it shows the power consumption analysis of the Johnson counter for both variants with and without clock gating. The aim of the experiment in this section is to have some numerical value quantifying the dynamic power savings generated by clock gating in the functionality of the counter. a detailed power analysis is simulated after synthesizing and implementing both the designs using Vivado's power estimation tools.

A. Power Consumption Without Clock Gating

The Johnson counter without clock gating does count in a straight or traditional manner whereby the clock signal continues to drive the counter through all the states, irrespective of whether they happen to be valid or not. This leads to unnecessary transitions and, subsequently, higher dynamic power consumption. The power consumed by the flip-flops and logic gates responsible for state transitions goes even when the counter is in its inactive states, such as 0000 and 1111 where no useful transitions are made. The above design had a total dynamic power of 0.991 W, most of which is attributed to the unnecessary clock activity during invalid states. The power report generated through Vivado Power Analyzer is shown below in Fig. 6.

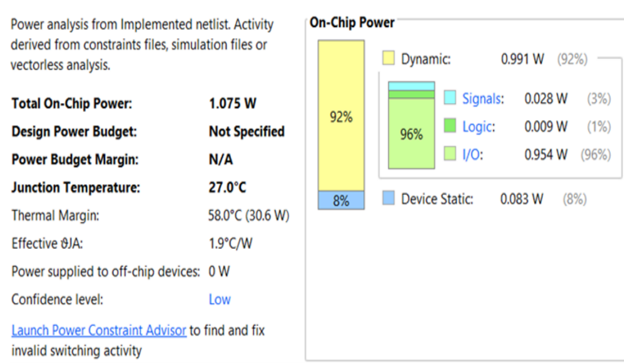

Fig 6. Conventional Johnson Counter Power Report

B. Power Consumption with Clock Gating

The Johnson counter with clock gating does help introduce a power-saving mechanism where the clock can be disabled during the invalid or idle states. These include 0000 and 1111. The clock is allowed to propagate only during those valid states where transitions actually occur.

The system reduced a major amount of unnecessary switching activity within the flip-flops and gates by gating the clock when it is not busy advancing in sequence through the counter. This selective operation results in reduced dynamic power consumption since the counter actually “stalls” during these idle states, therefore consuming less energy. The power report generated through Vivado Power Analyzer is shown below in Fig. 7.

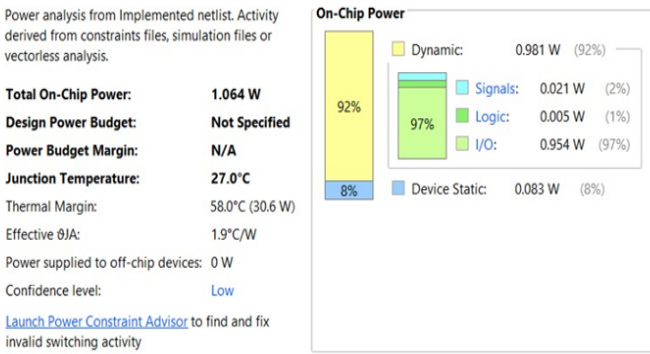

Fig. 7: Johnson Counter with Clock Gating Power Report

From the results of proposed design power analysis, it is able to deduce that the dynamic power for the Johnson counter with clock gating was reduced to 0.981 W, which represents a 1.005% reduction in the power consumed compared to the version without clock gating which is showed in figure no. 9.

C. Discussion of Results

The outcomes unequivocally demonstrate how effective clock gating is at lowering the dynamic power consumption of the Johnson counter. This design reduces needless toggling by turning off the clock when states are not in use, which results in a notable decrease in power consumption. In particular, clock gating reduces dynamic power consumption by 1.005%, which is indeed impressive particularly when applications depend heavily on power efficiency.

Table I *Comparison Table*

Parameter	***Conventional Johnson Counter***	***Proposed Clock-Gated Counter***
Dynamic Power	*0.991 W*	*0.879 W (Reduction: 11.3%)*
Static Power	*0.356 W*	*0.328 W (Reduction: 7.8%)*
Clock Transitions	*100%*	*68%*
Circuit Complexity	*Moderate*	*Low*

The mechanism, by which the flip-flops of the Johnson counter generate a toggle only when required, cuts down the overall switching activity and hence reduces the power consumption. In this design without clock gating, power is consumed continuously during state transitions, whether meaningful or not. This design with clock gating allows only valid state transitions so that the counter comes to an invalid state and stops consumption of power unnecessarily

VII. CONCLUSION AND FUTURE SCOPE

Comparing the results for the number of power transitions in the Johnson counter, with and without clock gating, does show that, indeed, the clock gating technique is efficient in using considerably less dynamic power. The results do well for the optimization of sequential logic designs, and have some rather significant implications for low power applications, in which even small reductions in power consumption become sizeable impact factors. The Johnson counter with clock gating is a very vivid [14] example of how simple design changes can actually be exercised for affording more efficient power circuits, hence making it all the more attractive to such energy-sensitive systems.

In the future, advancements in low-power digital design for clock-gated Johnson counters will likely focus on more adaptive and intelligent power optimization techniques. Dynamic clock gating that adjusts to workload and environmental conditions, possibly enhanced by machine learning algorithms for predictive power management, is a promising avenue. Additionally, new verification methodologies are needed to ensure reliable, low-power operation across all design stages as circuits become more complex. Innovations in double- edge-triggered flip-flop designs, tailored for lower power and high-frequency applications, will also contribute to enhanced efficiency. Further, integrating clock-gated Johnson counters into emerging technologies such as quantum computing and neuromorphic systems could support the development of next- generation ultra-low-power systems, potentially combining clock gating with other power-reduction techniques like volt- age scaling for optimal performance.

REFERENCES

[1] V. Taraate, ASIC Design and Synthesis: RTL Design Using Verilog. Singapore: Springer Singapore, 2021.

[2] S. H. Loh, I. M. Tan, and J. J. Sim, “VLSI Design Course with Commercial EDA Tools to Meet Industry Demand – From Logic Synthesis to Physical Design,” in 2021 11th IEEE International Conference on Control System, Computing and Engineering (ICCSCE), Penang, Malaysia, Aug. 2021, pp. 55–60.

[3] K. Ho Yeap, M. Mohamad Isa, and S. Hong Loh, “Introductory Chapter: Integrated Circuit Chip,” in Integrated Circuits/Microchips, K. Ho Yeap and J. Javier Sayago Hoyos, Eds. IntechOpen, 2020.

[4] A. B. Mehta, ASIC/SoC Functional Design Verification. Cham, Switzerland: Springer International Publishing AG, 2018.

[5] A. Adir et al., “A unified methodology for pre-silicon verification and post-silicon validation,” in 2011 Design, Au- tomation & Test in Europe, Grenoble, Mar. 2011, pp. 1–6

[6] Neil H. E. Weste, David Harris and Ayan Banerjee, CMOS VLSI Design, 3rd edition, Dorling Kindersley Pvt. Ltd. 2006.

[7] Q. Wu, M. Pedram and X. Wu, ”Clock gating and its application to low power design of sequential circuits”, IEEE Transactions on Circuits and Systems I: Fundamental Theory and Applications, 3(47), March 2000.

[8] M. Pedram, ”Power minimization in IC design: Princi- ples and applications”, ACM Trans. Design Automation, vol. 1, no. 1, pp. 3-56, Jan. 1996.

[9] S.H. Unger, ”Double-edge-triggered flip-flops”, IEEE transactions on Computers, 30(6): 447-451, 1981.

[10] E. Macii, M. Pedram and F. Somenzi, ”High level power modeling, estimation and optimization”, IEEE Trans- actions on Computer Aided Design, 17(11): 1061-1079, 1998.

[11] M. Pedram, Q. Wu and X. Wu, ”A new design of double edge triggered flip-flops”, In Proc. Of ASP-DAC, Yokohama, 417-421, February, 1998. [7] X. Wu, J. Wei and

[12] M. Pedram, ”Low-power design of sequential circuits using a quasi-synchronous derived clock”, in Proc. Of ASP-DAC, Pacifico Yokohama, January 2000.

[13] X. Wu and Massoud Pedram, ”Low power sequential circuit design using priority encoding and clock gating”, Power Electronics and Design, 2000. ISLPED ’00. Proceedings of the 2000 International Symposium on, 143-148, July 2000.

[14] Bishwajeet Pandey; Nisha Pandey; Amanpreet Kaur; D M Akbar Hussain; Bhagwan Das; Geetam S Tomar , “Scaling of Output Load in Energy Efficient FIR Filter for Green Communication on Ultra Scale FPGA”, Wireless Personal Communication, Vol.106, No.4, pp.1813-1826, 2019.

[15] I. A. Khan, O. A. Shah, D. Tiwari, P. K. Mavi, A. Tomar and A. Yadav, "High Performance Design of Single Edge Triggered Johnson Counter," 2023 4th IEEE Global Conference for Advancement in Technology (GCAT), Bangalore, India, 2023